REVISION QUESTIONS FOR SA220-SUSTAINABLE AGRICULTURE 2

QUESTION 1

a) You have been invited by Mrs. Tembo of Chibombo district to make a farm visit and do an assessment for earthen pond aquaculture development. Explain three main factors which you farm assessment would focus on. **[3 Marks]**

b) Assuming the farm is suitable for earthen pond aquaculture, what type of fingerlings would you recommend her to procure if the aquaculture is to be done for business and why? **[2 Marks]**

c) Mrs Tembo plans to be using maize bran as a source of nutrition for the fish. Based on your understanding of fish feed and the main reasons of feeding fish, what would be your advice to Mrs Tembo. [**5 Marks**].

d) She has been doing personal research on YouTube, she came across some technical words that she needs further information on. Please help her to understand the following terms by defining or differentiating them. [**5 Marks**]

   i. BROODSTOCK

   ii. HYPOPHYSATION

   iii. MONOSEX CULTURE

   iv. FISH AND FISHES

   v. INGREDIENT

QUESTION 2

a) List and explain four importance of aquaculture to a country of your choice [**8 Mark**]

b) Explain the main functions of the principal components of a fish cage [**10 Marks**]

c) List and explain two challenges of that come with cage fish farming [**2 Marks**]

QUESTION 3

a) With examples where possible, discuss the factors to be considered when selecting fish species for aquaculture [**6 Marks**).

b) In aquaculture, understanding puberty of your fish species is critical because both Early and Late puberty can be a problem. With examples, discuss the previous statement [**8 Marks**]

c) Explain two common reproductive dysfunctional in male fish [**4 Marks**]

d) Give two drawbacks of using hormones as a way of correcting the reproductive dysfunctional of fish species [4 Marks]

Question 5

a) Aquacultures development is dependant on various human interventions in the normal reproductive processes of fish. One such key intervention is artificial propagation. Explain three advantages of artificial propagation [**6 Marks**]

b) You have received a call from the Airport to assist Immigration Officers to identify the gender of catfish suspected to be imported illegally in the country. Kindly, explain three ways of identifying the sex of the species in question. [**6 Marks**]

c) Using your understanding of what aquaculture is, kindly explain four ways in chich it can be classified [**8 Marks**]

QUESTION 6

a) Production of mono-sex tilapia is done in several ways, each having advantages and disadvantages. Discuss five ways of producing mono-sex tilapia giving merits and demerits for each [**10 Marks**]

b) Which sex of tilapia is preferred for aquaculture and why? [**10 Marks**]

QUESTION 7

a) Fish hatcheries occupy a strategic position in the advancement of sustainable aquaculture development in any country. Discuss the roles that fish hatcheries play in the aquaculture industry [**8 Marks**]

b) What are the characteristics of a good broodstock [**8 Marks**]

c) It is a common practice to starved fingerlings for at least 24hour before they are transported to for stocking at another fish farm. What are the advantages of this hatchery practice [**4 Marks**]

QUESTION 8

a) Define biosecurity [**2 Marks**]

b) Discuss the main three principals under which aquaculture biosecurity operates [**6 Marks**]

c) List and explain the major ways of pathogen transmission [**10 Marks**]

d) Explain two benefits of biosecurity [**2 Marks**]

QUESTION 9

a) Formulating fish feed is not easy because there are a number of things which must be know if the feed is to make an impact on the fish. As such, you are asked to discuss any five types of information that is needed to properly formulate aqua-feeds? [**10 Marks**]

b) Discuss the advantages of complete feed to a species of your choice [**10 Marks**]

QUESTION 10

a) The life stage of fish determines the nutritional requirement for the fish. True or False? Defend your answer [**10 Marks**]

b) What are the main objectives of feed formulation [**6 Marks**]

c) A feed is only as good as its ingredients. Briefly, discuss the statement [**4 Marks**]

QUESTION 11

a) Nutritional requirements for fish vary from species to species and or fish to fish. Discuss four (4) factors that affect nutritional requirements of fish. [**8 Marks**]

b) There are six classes of nutrients that are needed by fish just like other animals. However, not all of the classes provide energy to the fish. Which of the classes provide energy to the fish and explain protein sparing concept in fish feed. [**8 Marks**]

c) Explain the four methods of delivering fish feed to fish [**4 Marks**]

QUESTION 12

a) Why is important from an aquaculture perspective to know or study reproductive diversity [**10 Marks**]

b) Discuss four (4) mating systems in fish [**8 Marks**]

c) Define the following terms [**2 Marks**]

   i. Oviparity

   ii. Viviparity

QUESTION 13

a) Discuss the four (4) types of fish mouth and how they relate to feeding and choice of feed types [**10 Marks**]

b) What are the advantages and disadvantages of manual and automatic feeding methods? [**10 Marks**]

QUESTION 14

a) Discuss the gains and losses of dissolved oxygen in water [**10 Marks**]

b) Explain ways of increasing dissolved water in a pond [**4 Marks**]

c) Explain pH fluctuations in pond culture [**6 Marks**]

QUESTION 15

a) A fish farmer is suspecting that the fish in the culture facility maybe be diseased. However, due to lack of knowledge and expertise is on the subject of fish diseases, you are requested to provide a brief write up of common fish behaviour that maybe a sign of disease in a production facility [**10 Marks**]

b) List the main causes of fish diseases [**5 Marks**]

c) The presence of a pathogen is the same as a disease. Discuss the statement [**5 Marks**]

QUESTION 16

a) Explain the fish disease triangle [**10 Marks**]

b) What would be the impact of Tilapia Lake Virus on the Zambian Aquaculture industry? [**10 Marks**]

QUESTION 17

a) Using the data in the following table 1, Calculate the Feed Conversion Ratio (FCR) for the three species and give guidance on which one of the three species (**1,2 and 3**) would not make a better species for farming and why. **(10 marks)**

Table 1: Data about three fish species

SPECIES	QUANTITY OF FEED ADMINISTERED (G)	WEIGHT GAINED (G)
1	4.8	3.1
2	5.2	2
3	10.7	9.7

b) Mr. Mwamba would like to establish a hatchery at Chilubi Island but he doesn't know where he could source broodstock for his hatchery. He has been informed that you took an aquaculture course as part of your training at KMU and he reaches out to you to provide guidance on this issue. What would you tell Mr. Mwamba are the main sources of broodstock [**6 Marks**]